

# PRISM: Past-Regularized Iterative Self-Distillation with Momentum for Polyp Segmentation

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## Introduction

Polyps are abnormal tissue growths in the colon and are considered early indicators of colorectal cancer. Early detection and accurate segmentation of polyps are therefore crucial for improving diagnosis and treatment outcomes. While deep learning-based segmentation models have shown promising results, they often suffer from limited generalization to unseen data and high computational cost.

We propose **PRISM** (Past-Regularized Iterative Self-Distillation with Momentum) to overcome these challenges. Our method leverages knowledge distillation in an iterative teacher-student framework with three key contributions:

- **Momentum-based teacher**, updated by Exponential Moving Average (EMA) [1] of student weights, which provides smoother parameter updates and prevents noisy fluctuations. This stabilizes the teacher and offers consistent guidance to the student.
- **Temperature-scaled sigmoid**, producing softer and more informative targets to regularize training.
- **Improved segmentation accuracy** without adding extra inference cost, making the method efficient in practice.

## Method

- Teacher parameters updated with EMA:

$$\theta_{teacher}^t = \gamma \theta_{teacher}^{t-1} + (1 - \gamma) \theta_{student}^t$$

(smooths teacher updates and stabilizes training)

- Temperature scaling smooths outputs:

$$\sigma_T(x) = \frac{1}{1 + e^{-x/T}}$$

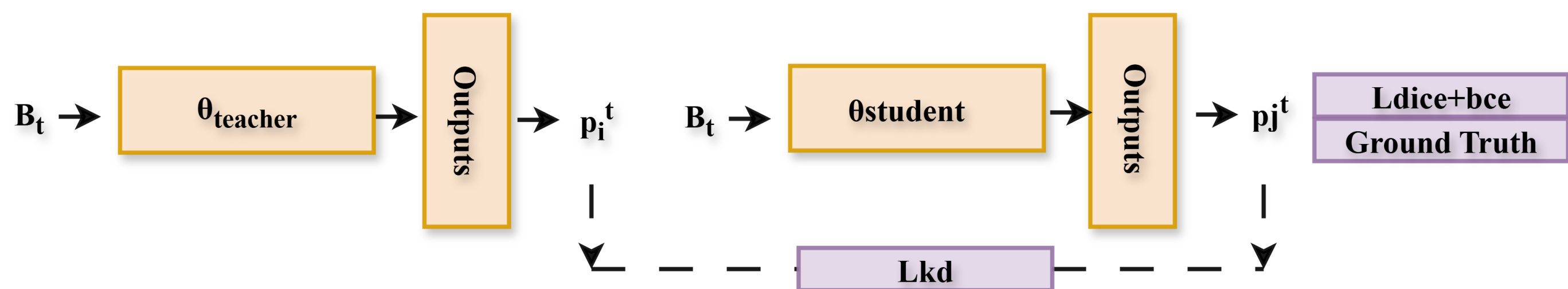
- Combined loss:

$$L = L_{Dice} + L_{BCE} + \lambda L_{KD}$$

- Losses:

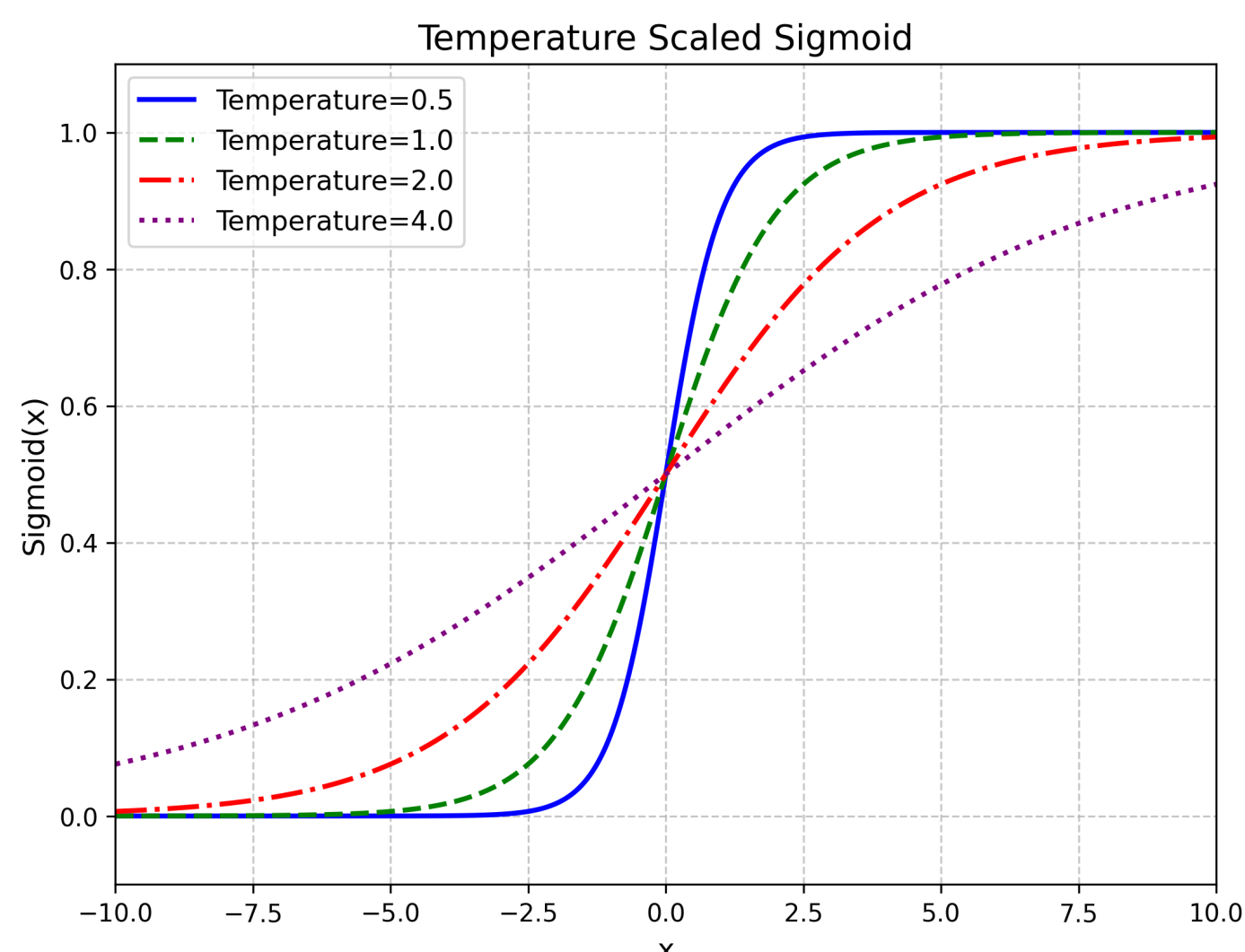
- $L_{Dice}$ : measures overlap between prediction and ground truth
- $L_{BCE}$ : pixel-wise binary cross-entropy
- $L_{KD}$ : MSE between student and teacher outputs

- Distillation starts after 6 epochs for stable feature learning.



Student and teacher networks receive the same input. The student is optimized using segmentation and distillation losses, while the teacher is updated indirectly through Exponential Moving Average of the student weights.

## Effect of Temperature Scaling



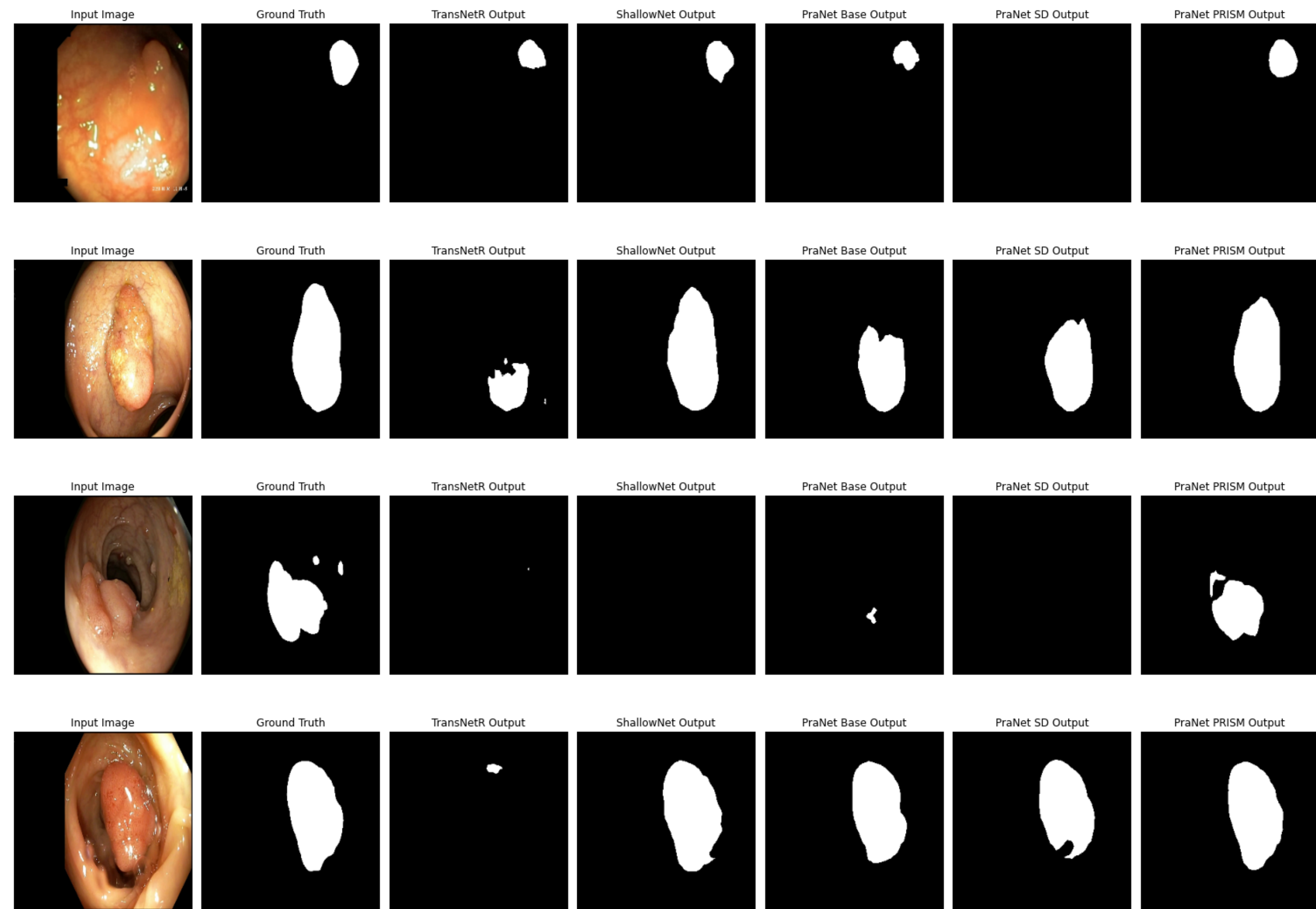
Effect of temperature scaling on the sigmoid function. Lower temperatures (e.g.,  $T = 0.5$ ) yield sharper, more binary-like transitions, while higher temperatures (e.g.,  $T = 4.0$ ) produce smoother outputs.

## Datasets

| Dataset | Images | Center                              |
|---------|--------|-------------------------------------|
| data c1 | 256    | Ambroise Paré Hospital (Paris)      |
| data c2 | 301    | Istituto Oncologico Veneto (Padova) |
| data c3 | 457    | IRCCS, Italy                        |
| data c4 | 227    | Oslo University Hospital (Norway)   |
| data c5 | 208    | John Radcliffe Hospital (Oxford)    |
| data c6 | 88     | Univ. of Alexandria (Egypt)         |

These datasets were collected from multiple international medical centers [2], ensuring diversity in patient populations and imaging conditions.

## Model Output Comparison



Comparison of model outputs from TransNetR, ShallowNet, PraNet, PraNet with SD, and PraNet with PRISM, highlighting their differences in segmentation performance on the data c6 dataset.

## Results

| Method                | Dice        | IoU         | Precision   | Recall      |
|-----------------------|-------------|-------------|-------------|-------------|
| FCN                   | 0.76        | 0.68        | 0.90        | 0.74        |
| U-Net                 | 0.63        | 0.55        | 0.76        | 0.66        |
| TransNetR             | 0.72        | 0.66        | <b>0.93</b> | 0.70        |
| ShallowNet            | 0.76        | 0.70        | <b>0.93</b> | 0.77        |
| HarDNet-MSEG          | 0.77        | 0.70        | 0.88        | 0.78        |
| PSPNet                | 0.80        | 0.72        | 0.88        | 0.79        |
| ResNetUNet(ResNet34)  | 0.79        | 0.73        | 0.92        | 0.78        |
| DeepLabV3+(ResNet50)  | <b>0.81</b> | <b>0.75</b> | 0.92        | 0.79        |
| PraNet                | 0.78        | 0.72        | 0.92        | 0.79        |
| PraNet + PRISM (Ours) | <b>0.81</b> | <b>0.75</b> | <b>0.93</b> | <b>0.80</b> |

Comparison of segmentation methods. Our PRISM-enhanced model achieves the best balance across all metrics.

## Ablation Study

| Method             | Dice        | IoU         | Precision   | Recall      |
|--------------------|-------------|-------------|-------------|-------------|
| TransNetR Base     | 0.72        | 0.66        | 0.93        | 0.70        |
| TransNetR + SD     | 0.74        | 0.67        | 0.91        | 0.73        |
| TransNetR + PRISM  | 0.76        | 0.69        | 0.90        | 0.77        |
| ShallowNet Base    | 0.76        | 0.70        | 0.93        | 0.77        |
| ShallowNet + SD    | 0.77        | 0.70        | 0.90        | 0.79        |
| ShallowNet + PRISM | 0.79        | 0.72        | 0.90        | 0.80        |
| PraNet Base        | 0.78        | 0.72        | 0.91        | 0.79        |
| PraNet + SD        | 0.80        | 0.74        | 0.93        | 0.80        |
| PraNet + PRISM     | <b>0.81</b> | <b>0.75</b> | <b>0.93</b> | <b>0.80</b> |

PRISM consistently surpasses both Base training and standard Self-Distillation (SD).

## Conclusion

- PRISM introduces EMA-based momentum teacher for stable supervision.
- Improves Dice and IoU across multiple backbones without inference cost.
- Robust under domain shift (multi-center datasets).
- Future work: adaptive momentum scheduling, hybrid loss functions.

## References

- [1] K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick, "Momentum contrast for unsupervised visual representation learning," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 9729–9738, 2020.
- [2] S. Ali, D. Jha, N. Ghatwary, S. Realdon, R. Cannizzaro, O. E. Salem, D. Lamarque, C. Daul, M. A. Riegler, K. V. Anonsen, *et al.*, "A multi-centre polyp detection and segmentation dataset for generalisability assessment," *Scientific Data*, vol. 10, no. 1, p. 75, 2023.